



Genome-wide analysis of spatiotemporal gene expression patterns during floral organ development in *Brassica rapa*

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Abstract

Flowering is a key agronomic trait that directly influences crop yield and quality and serves as a model system for elucidating the molecular basis that controls successful reproduction, adaptation, and diversification of flowering plants. Adequate knowledge of continuous series of expression data from the floral transition to maturation is lacking in *Brassica rapa*. To unravel the genome expression associated with the development of early small floral buds (< 2 mm; FB2), early large floral buds (2–4 mm; FB4), stamens (STs) and carpels (CPs), transcriptome profiling was carried out with a Br300K oligo microarray. The results showed that at least 6848 known nonredundant genes (30% of the genes of the Br300K) were differentially expressed during the floral transition from vegetative tissues to maturation. Functional annotation of the differentially expressed genes (DEGs) (fold change ≥ 5) by comparison with a close relative, *Arabidopsis thaliana*, revealed 6552 unigenes (4579 upregulated; 1973 downregulated), including 131 Brassica-specific and 116 functionally known floral Arabidopsis homologs. Additionally, 1723, 236 and 232 DEGs were preferentially expressed in the tissues of STs, FB2, and CPs. These DEGs also included 43 transcription factors, mainly AP2/ERF–ERF, NAC, MADS–MIKC, C2H2, bHLH, and WRKY members. The differential gene expression during flower development induced dramatic changes in activities related to metabolic processes (23.7%), cellular (22.7%) processes, responses to the stimuli (7.5%) and reproduction (1%). A relatively large number of DEGs were observed in STs and were overrepresented by photosynthesis-related activities. Subsequent analysis via semiquantitative RT-PCR, histological analysis performed with in situ hybridization of *BrLTP1* and transgenic reporter lines (*BrLTP* promoter::*GUS*) of *B. rapa* ssp. *pekinensis* supported the spatiotemporal expression patterns. Together, these results suggest that a temporally and spatially regulated process of the selective expression of distinct fractions of the same genome leads to the development of floral organs. Interestingly, most of the differentially expressed floral transcripts were located on chromosomes 3 and 9. This study generated a genome expression atlas of the early floral transition to maturation that represented the flowering regulatory elements of *Brassica rapa*.

Keywords *Brassica rapa* · Floral organs · cDNA microarray · Flower development · Differential gene expression · MADS transcription factors

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Soo In Lee and Muthusamy Muthusamy contributed equally to this study.

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Introduction

Brassica rapa ssp. *pekinensis* (Chinese cabbage) is a popular vernalization-sensitive leafy vegetable belonging to the Brassicaceae family and is widely known for its nutritional and economic importance (Jung et al. 2014). Presently, six Brassica species including *B. rapa* (AA, $2n = 20$) are

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cultivated worldwide. *Brassica rapa* is a long-day, biennial, mesopolyploid species displaying high morphological diversity (Bonnema et al. 2011), and different cultivars display different flowering times (Xiao et al. 2019). Flowering and its timing are considered key agronomic traits in leafy vegetables such as Chinese cabbage (*B. rapa* ssp. *pekinensis*), bok choy (*Brassica rapa* ssp. *chinensis*) and cabbage (*Brassica oleracea*) because those traits influence product value (Kim et al. 2007; Li et al. 2018). In addition to vernalization, flowering is also regulated by a network comprising six major pathways, including those related to endogenous signals (gibberellins and other phytohormones, aging, autonomy, and other genetic factors) and environmental signals (photoperiod, ambient temperature, abiotic stress such as drought) (Kim et al. 2007; Greenup et al. 2009; Srikanth and Schmid 2011; Krizek and Anderson 2013; Huang and Irish 2015; Song et al. 2016; Ryan et al. 2015; Huang et al. 2018). These regulatory signals are perceived, converged and integrated to initiate the conversion of vegetative stem primordia to floral primordia, which is known as floral transition/flowering. Flower development is a major developmental transitional phase in the life cycle of flowering plants and is considered a good model system for elucidating the molecular mechanisms that regulate flower development and determine successful reproduction, species diversity, crop yield and quality (Li et al. 2018).

Understanding the molecular basis of floral transition to maturation and other regulatory switches that control flowering in *B. rapa* cultivars is important for helping breed new/novel late-bolting lines of leafy *B. rapa* cultivars (Kakizaki et al. 2011). As inferred from flowering models of *A. thaliana*, the temporal and spatially restricted selective expression of distinct fractions of the same genome represents organogenesis of each floral organ. Molecular evidence has shown that, compared with other nonhomologous organs, developmentally related organs exhibit similar genome expression patterns (Ryan et al. 2015). Therefore, a genome-wide survey of continuous series of expression data from the earliest floral stages until maturation would provide an accurate understanding of the molecular basis underlying each organ/tissue development in plants (Dong et al. 2016). To date, many flowering genes/regulatory models have been elucidated in the model plant, *A. thaliana*. *B. rapa* and *A. thaliana* are close relatives and members of the Brassicaceae, and they diverged from a common hexaploid ancestor. Evidence has revealed that *B. rapa* underwent whole-genome triplication followed by extensive genome reorganization (fusions and/or fissions), homologous gene loss, and segmental duplication via the process known as fractionation, which led to the production of subgenomes and new functions (Sankoff et al. 2010). Polyploidization and subsequent fractionation events led to diploid *Brassica rapa* with a relatively larger genome than that in *Arabidopsis*

(Flagel and Wendel, 2009). In comparison with *Arabidopsis*, diploid *Brassica rapa* encodes 18% of *Brassica rapa*-specific flowering genes (Song et al. 2016). Nonetheless, the *Brassica* genus shares a significant syntenic relationship with *A. thaliana* at both the chromosome and gene levels (Tang et al. 2012; Town et al. 2006; Wang et al. 2015), mainly due to the preferential retention of key genes that are involved in developmental processes similar to those of ancestral genomes (Xiao et al. 2013). Similarly, genes associated with flower development in *B. rapa*, such as *CONSTANS* (*CO*)-like, *FLOWERING LOCUS T* (*FT*), *FLOWERING LOCUS C* (*FLC*) and *LEAFY* genes (Xiao et al. 2013; Song et al. 2016); male fertility (Dong et al. 2013); and vernalization-related genes (Li et al. 2018) were selectively retained, and most of them have been functionally described in *A. thaliana* (Zhang et al. 2018). In another study, it was shown that some of the male fertility-related genes of *B. rapa* were different from those of *A. thaliana* (Dong et al. 2013), indicating a possible difference in the flowering mechanism between those species.

Therefore, adequate and systematic investigation is needed to understand the importance of the spatiotemporal expression patterns of several key and unique floral genes, the important cues of floral transition, gametogenesis and other floral organ development in *Brassica* cultivars, including Chinese cabbage. With the availability of the *B. rapa* (Chiifu-401) genome sequence (Wang et al. 2011), microarray-based transcriptome profiling across floral tissues is a preferred option because of its simplicity, high throughput and reproducibility (Paritosh et al. 2018). Microarrays have been successfully used to determine the expression of pistil genes and their association with self-incompatibility in *B. campestris* (Kwun et al. 2004), inflorescence development in *B. napus* (Huang et al. 2014), the genes responsible for rapid flowering in *B. rapa* (Song et al. 2016), the bolting genes in Chinese cabbage (Dong et al. 2016), the impact of temperature and photoperiod on flowering time and the role of MADS-box genes in organ identity (Saha et al. 2015). Nevertheless, information concerning differential gene expression patterns in floral buds, namely, a continuous series of expression data from the earliest floral stages until maturation, is lacking in *B. rapa*.

To fill this knowledge gap, we employed the highly reliable Br300K microarray chip covering 47,548 unigenes (Dong et al. 2013) to measure the genome-wide gene expression patterns in small early flower buds (FB2), large early flower buds (FB4), stamens (STs) and carpels (CPs) during the floral transition. Moreover, the tissue specificity and expression patterns of differentially expressed genes (DEGs) were validated via semiquantitative reverse transcription (RT)-PCR, RNA in situ hybridization and transgenic reporter lines (promoter::GUS). This microarray-based systematic identification of candidate genes during the early

floral transition to maturation generated a reference dataset for stage-specific gene expression and provided an important resource for future genetic manipulation towards the development of high-quality leafy vegetables.

Materials and methods

Plant growth conditions, vernalization and RNA isolation

Chinese cabbage (*B. rapa* ssp. *pekinensis*, inbred line ‘Chiifu’) plants were grown under greenhouse conditions (22/18 °C, 16/8 h light/dark photoperiod) for 4 weeks. For vernalization, the plants were transferred to a 4 °C cold chamber for 7 weeks and then transferred to a greenhouse for flowering. For RNA preparation and microarray analysis, tissues were collected from the vernalized small (<2 mm) early floral buds (FB2), large (2–4 mm) early floral buds (FB4), stamens (STs), carpels (CPs), and nonvernalized leaves (4 weeks old) as controls (CON). For RT-PCR analysis, additional samples such as vernalized leaves, leaves maintained at optimal conditions after vernalization (LAV), open flowers and siliques were collected. The samples were immediately frozen in liquid nitrogen prior to total RNA preparation. Total RNA was extracted using an RNeasy Plant Mini Kit (Qiagen, Germany) according to the manufacturer’s instructions, and the quantity and quality were measured with a spectrophotometer.

Floral transcriptome profiling by Br300K microarrays

A *B. rapa* Br300K (previously KBGP-50 K) oligo microarray chip that consisted of 331,384 oligo probes (60-mer) covering the nucleotide sequences of 47,548 unigenes (Kim et al. 2010; Dong et al. 2013) (NCBI Gene Expression Omnibus platform accession No. GPL17248; Online Resource S1) was used. The microarray analysis was repeated twice (for the FB2, FB4, ST and CP samples) or thrice (for CON samples) using the total RNA of biologically independent samples. The obtained microarray data were normalized and processed as described previously (Lee et al. 2008; Dong et al. 2013; Jung et al. 2014). The complete genome-wide expression data, including raw data from this study, were submitted to the Gene Expression Omnibus database (<https://www.ncbi.nlm.nih.gov/geo/>) under accession number GSE128989. Probe intensity (PI) (the average perfect match values of probes for each unigene) values were ultimately used as expression values of unigenes, and unigenes with a $PI \geq 1000$ (adjusted $P < 0.05$) were selected. The fold change was calculated against the mean PI values of the CON, and

those unigenes with > 2 or fivefold changes in expression were considered potential flowering candidate genes.

Functional annotation

The DEGs from the microarray data were annotated using the *B. rapa* gene set (version 2.5 [released on 28 Nov. 2017]) predicted by Wang et al. (2015) and Cai et al. (2017). The gene sequences retrieved from the *B. rapa* genome database (<http://brassicadb.org/brad/>) and the nucleotide and amino acid sequence homology were analyzed using BLASTN and BLASTX algorithms. For stringent annotations, the BLASTN and BLASTX cut-off values were set to $1E-5$ and $\geq 99\%$ sequence identity for a given query sequence containing at least 100 nucleotides or 30 amino acids. To identify homologous genes, the query sequences were also searched against the NCBI nonredundant database and TAIR10 protein database (<https://www.Arabidopsis.org/>).

In addition, conserved protein domains and Gene Ontology (GO) terms were assessed with InterProScan (ver. 5.28-67.0, <https://www.ebi.ac.uk/interpro/>) and Blast2GO (ver. 5.1.12, <https://www.blast2go.com/>) programs. GO enrichment analysis was performed using Fisher’s exact test in Blast2GO for DEGs with a $\geq$ twofold expression change, with $P < 0.05$ and an FDR < 0.05 . Transcription factors (TFs) and transcriptional regulators (TRs) were classified using iTAK standalone software (ver. 1.7, <http://itak.feila.b.net/cgi-bin/itak/index.cgi>). The DEGs were also compared with genes associated with flowering time, meristem identity and flower development in Arabidopsis (Bouché et al. 2015) to identify homologous flowering genes.

Semiquantitative RT-PCR analysis

Primers specific to seven DEGs (Brapa_ESTC000813, Brapa_ESTC001844, Brapa_ESTC003664, Brapa_ESTC008221, Brapa_ESTC010699, Brapa_ESTC012118 and Brapa_ESTC017129) were synthesized (Online Resource S2) for RT-PCR. Primers specific to the *B. rapa* actin gene sequence, *BrActin1* (EX087730), were used as a control. Total RNA (1 µg) extracted from floral/reproductive tissue (FB2, FB4, ST, CP, floral buds 2–3 mm in length, open flowers, siliques) and vegetative tissue [CON, 5 day-vernalized leaves (VL5), leaves collected at 5 days after vernalization (LAV5), and leaves collected at 10 days after vernalization (LAV10)] using an RNeasy Mini Kit (Qiagen, Germany) was reverse transcribed into first-strand cDNAs using an RNA to cDNA EcoDry Premix Kit (Clontech, USA) with oligo (dT) primers. 1 µl of cDNA was used as a template for PCR amplification with the following program: 94 °C for 5 min; 35 cycles of 94 °C for 1 min, 54 °C for 1 min, and 72 °C for 1 min; and a final extension of 72 °C for

7 min. The PCR products were resolved by electrophoresis on a 1.3% (w/v) agarose gel.

Phylogenetic relationships of key floral genes

To understand the evolutionary relationship of *SPL11*, *AGL42*, *NAC3*, *LTP1*, and *PISTILLATA* as well as some of the most abundant and functionally known genes, such as *WUS*, *SEP2*, and *APETALA*, the respective amino acid sequences of those genes from *A. thaliana*, *B. napus*, *B. oleracea*, *B. campestris*, *B. juncea*, *Capsella rubella*, *Camelina sativa*, *Raphanus sativus* and *Eutrema salsugineum* and other species (Online Resource Fig. 1) were retrieved from a protein database (<https://www.uniprot.org>) and aligned by a multiple sequence alignment algorithm using Clustal W. An unrooted phylogenetic tree was constructed with 1000 bootstrap values in conjunction with the maximum likelihood method using MEGA software (version 7.0.26; Kumar et al. 2016).

Localization and spatiotemporal expression patterns of LTP1 by RNA in situ hybridization

To confirm the accuracy of the expression patterns revealed by the microarray analysis of floral organs, *BrLTP1* (Brapa_ESTC001844) was selected for in situ hybridization. The cDNA fragments (300–600 bp) of *BrLTP1* were amplified with specific primers (Online Resource S2) and subcloned into a TA cloning vector, PCR-II-TOPO (Invitrogen). To prepare the probes, the recombinant plasmids were linearized, transcribed (in vitro), labeled using a digoxigenin SP6/T7 labeling kit (Roche Diagnostics) and then hydrolyzed to obtain 150–200 nucleotide-long fragments. Floral buds and open flowers were fixed under agitation at 4 °C for 12 h in phosphate-buffered saline (PBS) (1.3 M NaCl, 70 mM Na₂HPO₄, 30 mM NaH₂PO₄) containing 4% (w/v) paraformaldehyde, 0.1% (v/v) Triton X-100, and 0.1% (v/v) Tween-20 and then washed twice with PBS buffer for 30 min. The washed samples were dehydrated at room temperature via an ethanol series. The fixed samples were then embedded in paraplast (Sigma-Aldrich) and sectioned (8 µm each) using a microtome. The sections were deparaffinized and hydrated under RNase-free conditions on Probe-on Plus slides (Fisher Scientific). The sections were subsequently equilibrated in Tris–EDTA buffer (100 mM Tris (pH 8), 50 mM EDTA) at 37 °C and then treated with proteinase K (1 µg/mL) in Tris–EDTA solution for 30 min at 37 °C. Afterward, they were treated with 4% paraformaldehyde in PBS (pH 7) for 10 min (to fix) and then deacetylated with 0.1 M triethanolamine and acetic anhydride (pH 8) for 10 min before being dehydrated via an ethanol series. The tissue sections were hybridized with the probes at 48 °C for 16 h in a hybridization solution and then washed with solutions containing

2× SSC, 1× SSC, and 0.1× SSC for 15 min at 50 °C. The hybridized probes were detected colorimetrically using anti-DIG conjugated alkaline phosphatase (Roche, Germany) and visualized (Fig. 1) with a bright-field microscope combined with a digital camera (Zeiss, Germany).

Construction of BrLTP promoter::GUS expression cassettes, genetic transformation and GUS staining assays

The BrLTP promoter (1537-bp DNA fragment upstream of the transcription start site) was amplified (forward, 5'-AGG ATCCAATAGAGAGCTGCAGACATC-3'; reverse, 5'-AAC TAGTTTTTGGAATTAATTTGTAGTTC-3') from the genomic DNA of *B. rapa* ssp. *pekinensis* using an EX-Taq Polymerase Kit (Takara, Shiga, Japan). The PCR program was as follows: 95 °C for 5 min; 35 cycles of 95 °C for 30 s, 55 °C for 30 s, and 72 °C for 2 min; and then 72 °C for 10 min. The amplicons were cloned into a pGEM-T Easy vector (Promega, Madison, WI, USA) and then confirmed by Sanger sequencing. The elements of the promoter sequences were further analyzed via the plantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) and PLACE (<http://www.dna.affrc.go.jp/PLACE/>) databases. The promoter region was further cloned into the *Bam*HI and *Spe*I sites of a pCambia1381 expression vector, and the final construct was introduced into Chinese cabbage DH03 by *Agrobacterium tumefaciens* (strain GV3101)-mediated transformation, as described by Kim et al. (2007). The floral buds and flowers from transgenic lines (T₂ generation) expressing the BrLTP promoter::GUS construct were screened for GUS activity according to the methods of Jefferson et al. (1987). The GUS expression in floral tissues was digitized (Fig. 2b) using an Olympus SZX12 stereomicroscope (Olympus, Tokyo, Japan).

Results

To determine the genome expression associated with the floral transition and generate a reference data set for the organ-specific transcriptome of *B. rapa*, we profiled the transcriptomes of early flower buds (FB2, FB4), STs and CPs and their expression patterns using a Br300K microarray, which covered 47,548 unigenes of Brassica species. The array included 26,394 known unigenes [representing 19,482 nonredundant genes; 43% of the total gene models of *B. rapa* (Zhang et al. 2018)] and 21,154 unigenes representing predicted genes. Approximately 85% of the unigenes (40,528) were found to exhibit homology with Arabidopsis genes (18,295 gene models). Homology-based functional annotation with all other species showed that 90.5% of the unigenes could be annotated (Online

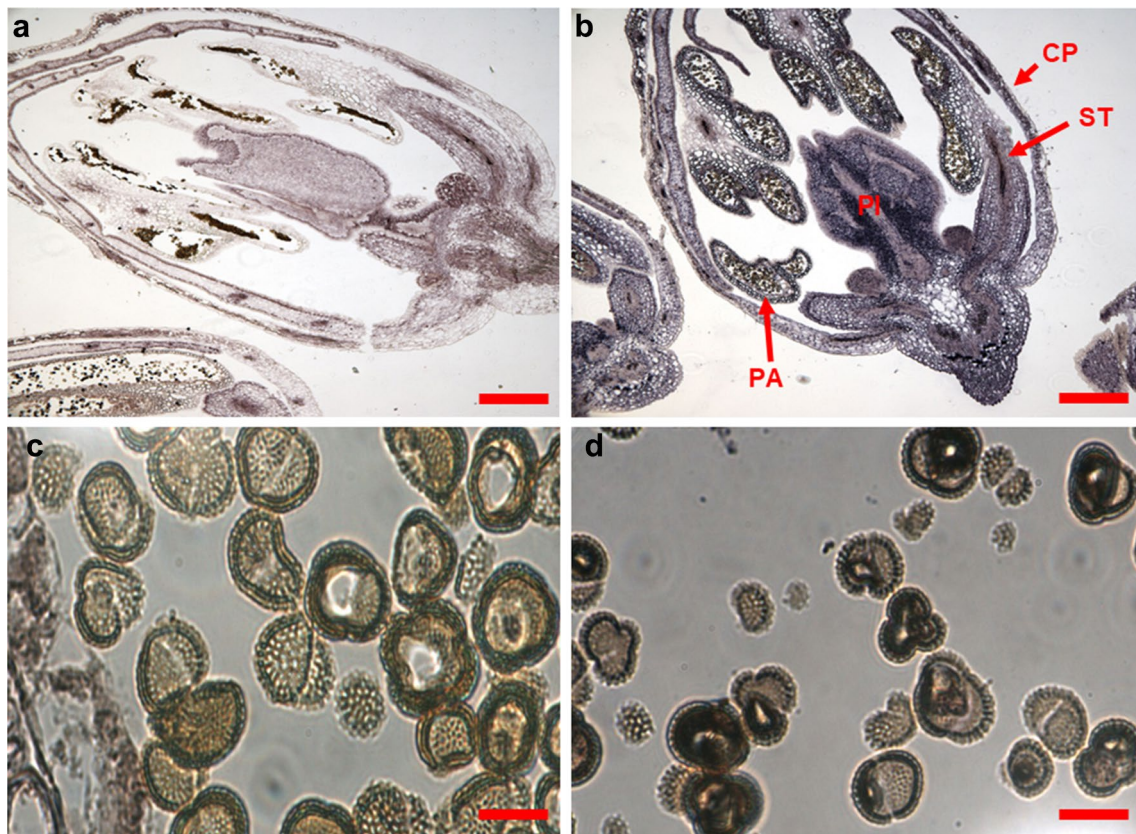


Fig. 1 In situ hybridization of *BrLTP1* (Brapa_ESTC001844) from floral sections (8 μ m thickness) of *Brassica rapa*. Cross-sections of *B. rapa* flowers were hybridized with digoxigenin-labeled sense (a and

c) and antisense (b and d) probes of *BrLTP1*. A positive hybridization signal was observed in the carpels (CPs), pollen sac (PA), pistil (PI) and stamens (STs). Red bar = 200 μ m

Resource S1), suggesting 9.5% of the unigenes were predicted to be unique for *B. rapa* gene models. In the present study, 15,733 unigenes showed significant ($P < 0.05$) expression changes of more than twofold during the floral transition and subsequent floral organ development compared with those in the CON. However, 14,233 known and predicted differentially expressed unigenes (with a $PI > 1000$) accounting for 6848 gene models of *B. rapa* (approximately 30% of the total unigenes) were selected for consideration as flowering gene models (Online Resource S3). Interestingly, most of the DEGs originated from chromosomes 3 and 9 of the *B. rapa* genome. The expression patterns and/or magnitudes of expression level were different between the organs and corresponded to their developmental relatedness. To measure the reproducibility of the genome expression profiles, correlation coefficient values were calculated between technical replicates, and values ranging from 0.87 to 0.98 were considered for further analysis (Online Resource S4). Moreover, a comparison of the correlation coefficients (0.55–0.80) of the genome expression between floral tissues (FB2, FB4, and STs) and vegetative tissues (CON) indicated the true variation in expression patterns. In terms of quantitative

measurements, the smallest change in genome expression (correlation coefficient = 0.82–0.89) was observed between the CPs and CON.

Among the DEGs, 6552 (13.8% of total DEGs) showed a fivefold change in genome expression between floral and vegetative tissues (Online Resource S5). The development of FB2 and FB4 was associated with 3639 and 3929 DEGs, of which the expression of 2501 and 2735 was upregulated, respectively, and of which the expression of 1138 and 1194 was downregulated, respectively. Similarly, 5628 (3666 upregulated; 1962 downregulated) and 2314 DEGs (1962 upregulated; 827 downregulated) controlled the development of STs and CPs, respectively. A subset of 1536 DEGs were observed in all floral tissues; these DEGs are possibly involved in key cellular and metabolic processes associated with the floral transition and subsequent flower developmental processes (Fig. 3a). The overall functional assessment of 3755 DEGs (FC = 5) with known Gene Ontology (GO) terms (Online Resource S6) revealed that the floral transition was enriched with genes responsible for biological processes such as metabolic processes (23.7%) and cellular processes (22.7%), followed by responses to stimuli (7.5%) and reproduction

Fig. 2 Histological GUS staining assay of transgenic reporter lines (BrLTP promoter::GUS) of *Brassica rapa* L. ssp. *pekinensis*. **a** Simplified vector map of the expression of the BrLTP promoter::GUS construct in *B. rapa*. Promoter, promoter sequence of BrLTP; P35S, CaMV 35S promoter; T35S, terminator sequence of the CaMV35S gene; Tnos, terminator sequence of the nopaline synthase gene; *hph*, hygromycin phosphotransferase gene for selection of transgenic plant; HIS, histidine tag. **b** Digitalized GUS staining assay showing the expression of the BrLTP promoter::GUS in floral tissues of *B. rapa* transgenic lines

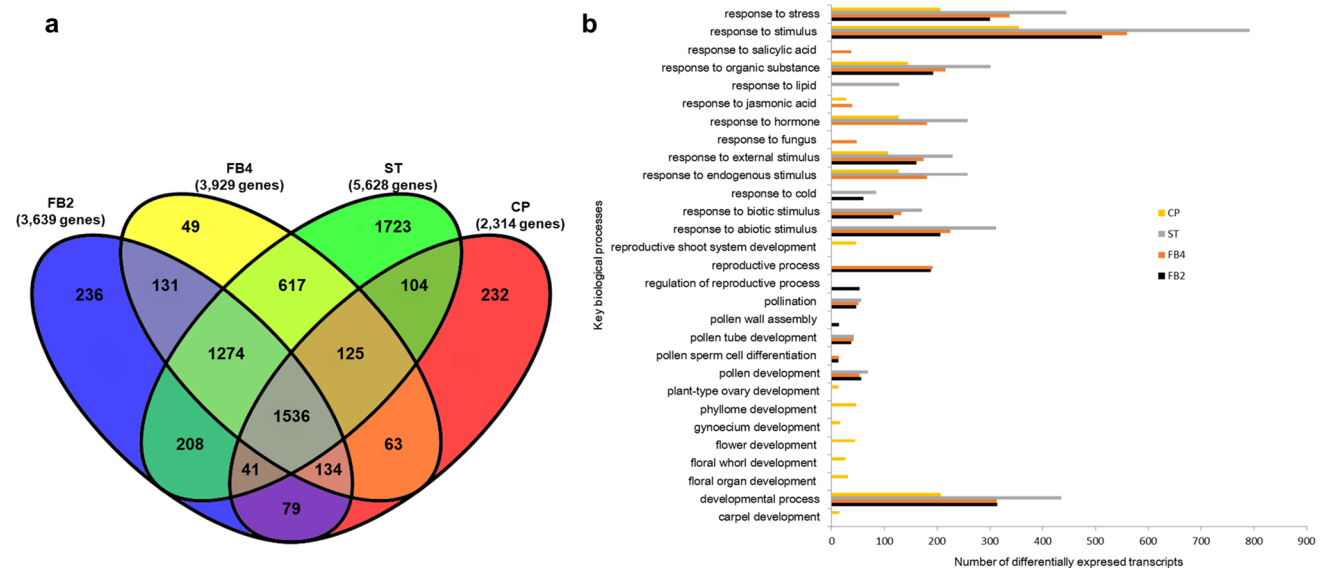
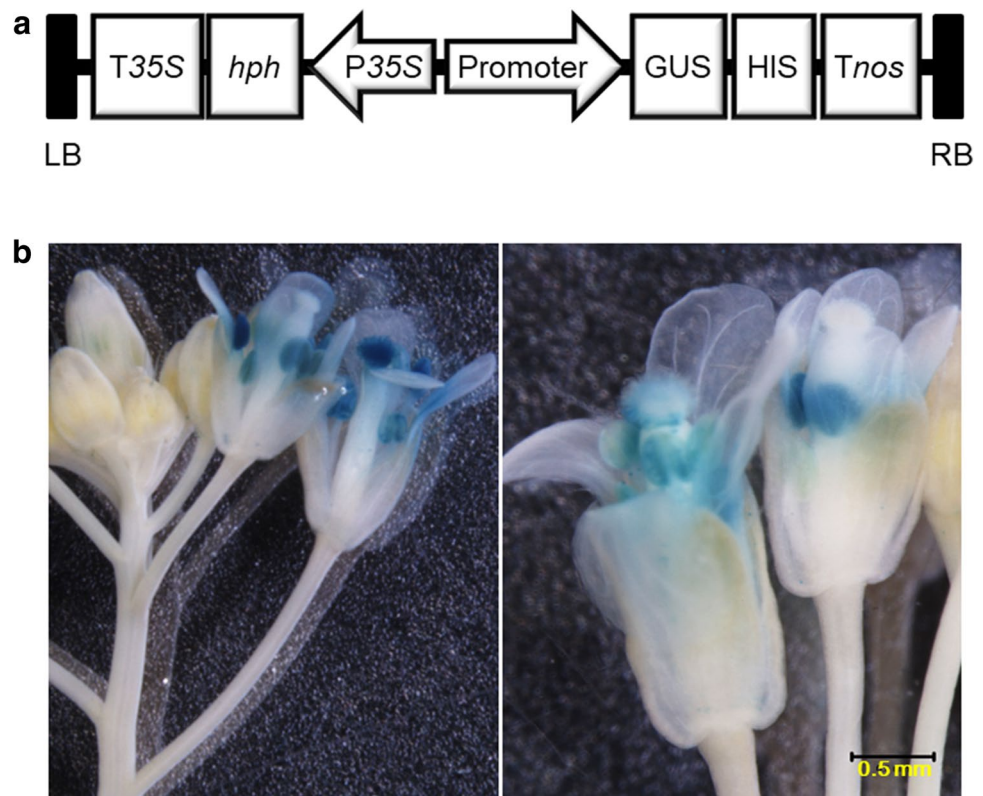


Fig. 3 Transcriptomic changes of floral transcripts (fold change ≥ 5 ; $P < 0.05$ and FDR < 0.05) in *Brassica rapa* **(a)** and **(b)** their key biological processes according to known Gene Ontology (GO) terms. FB2, floral buds (< 2 mm); FB4, floral buds (2–4 mm); ST, stamen and CP, carpel

(1%) (Fig. 3b). Among the molecular function categories, DEGs associated with catalytic activities and cell parts (18.8%) and organelles (13.3%) of cellular component synthesis were abundantly expressed. Interestingly, the expression of most of the DEGs (4579 of 6552) was

induced, while that of 1973 was reduced during flower development. Based on their regulatory pattern, 6552 DEGs were grouped into nine clusters (Fig. 4). Among them, seven clusters (1, 3, 4, 6, 7, 8 and 9) had 4174 DEGs with an abundant change in expression, whereas

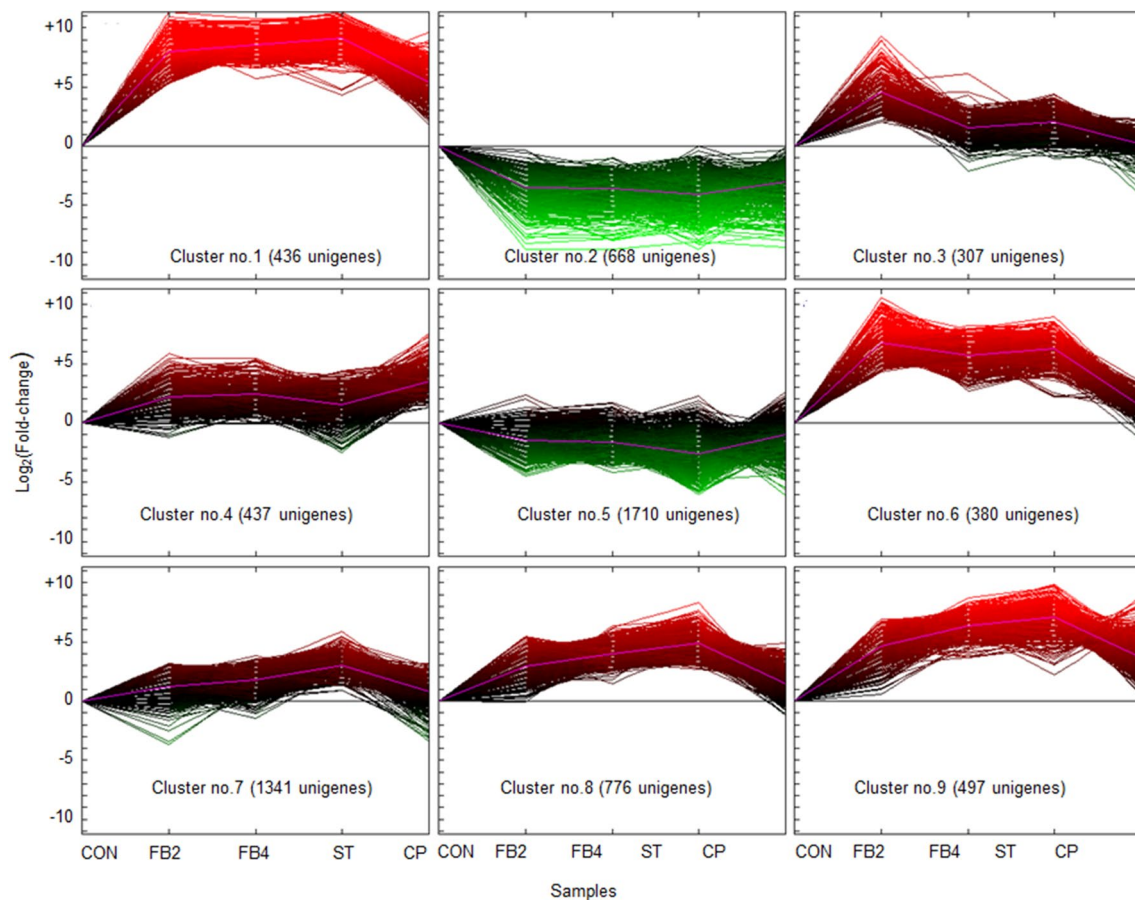


Fig. 4 K-means clustering of 6552 differentially expressed unigenes (fold change ≥ 5) during flower development in *Brassica rapa*. The unigenes were clustered according to their expression pattern using

the *K* means clustering method in TIGR MeV s/w. CON, leaf tissues; FB2, floral buds (<2 mm); FB4, floral buds (2–4 mm); ST, stamen; and CP, carpel

clusters 2 and 5 had 2378 DEGs with the least change in expression. Furthermore, 307 DEGs of cluster 3; 437 DEGs of cluster 4; and 2614 unigenes of clusters 7, 8, and 9 were preferentially expressed in FB2, CPs, and STs, respectively. The floral transition (FB2) was accompanied by the transcriptional reprogramming of no fewer than 3639 unigenes ($FC > 5$), where the expression of 1138 unigenes was downregulated and where the expression of 2501 was upregulated. The subsequent flower development is associated with both overlapping and stage-specific DEGs according to their organ relatedness. As expected, the highest number (3075) of overlapping DEGs was observed between the developmental stages of FB2 and FB4, while 2810 and 1536 overlapping DEGs participated in ST and CP development, respectively. Further analysis revealed that 1094 overlapping DEGs across four developmental stages were upregulated in comparison with those in the CON; these DEGs could be considered potential floral markers for *B. rapa*.

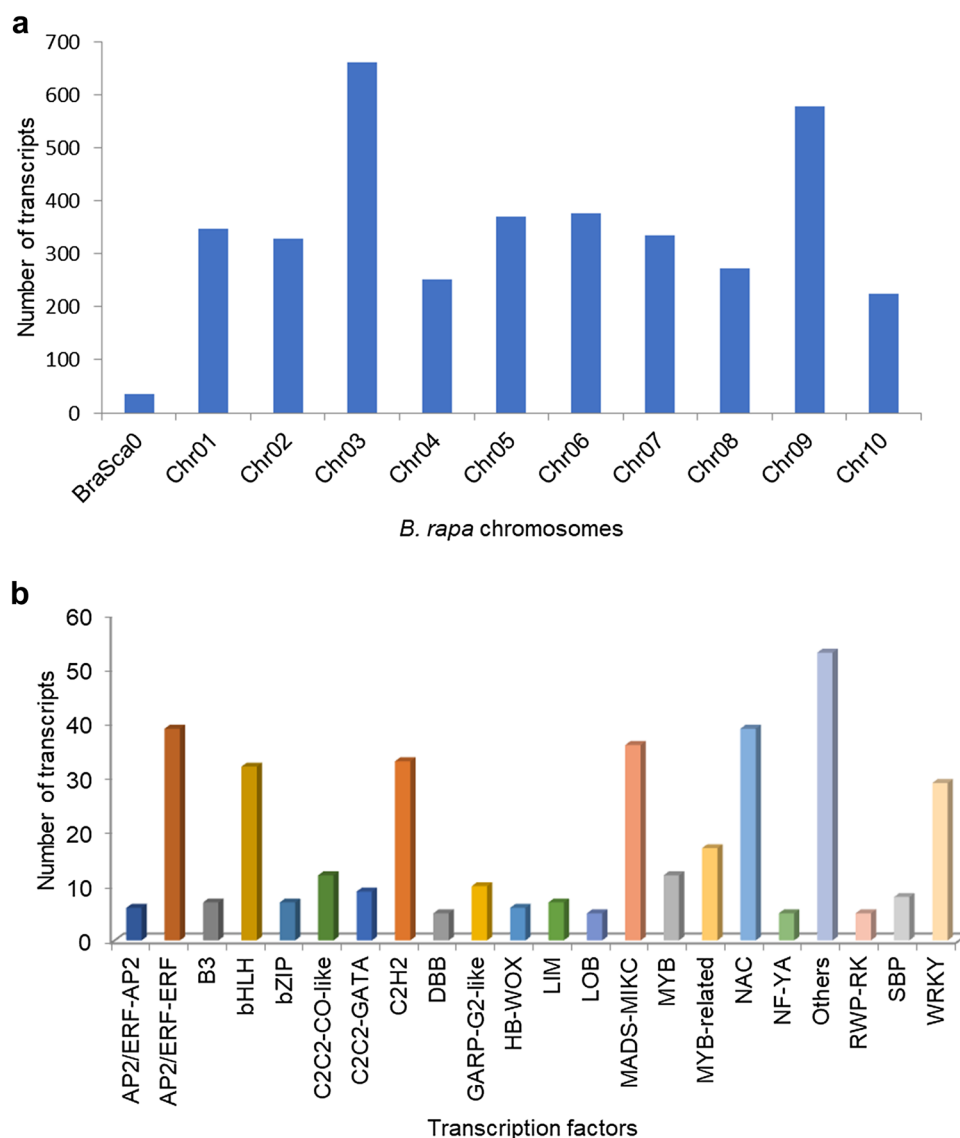
Overrepresentation analysis of DEGs ($FC = 5$) of each developmental stage and their functional assessment

To understand the importance of the differential genome expression of each floral organ or developmental stage during flower development, the overrepresentation of differentially expressed transcripts ($FC = 5$) of FB2, FB4, STs, and CPs was analyzed. The analysis of the genome expression of each organ provided some interesting facts on key biological, molecular and cellular processes regulating the floral transition, male and female reproductive organogenesis, and maturation. The most common and frequent gene expression changes detected in the floral organs were associated with transcripts responsive to stimuli (GO:0050896), stress (GO:0006950), and nitrogen compound metabolic processes (GO:0006807) in the biological process category; catalytic activity (GO:0003824), transcription regulator activity (GO:0140110), and oxidoreductase activity (GO:0016491)

in the molecular function category; and intracellular parts (GO:0044424), organelles (GO:0043226), intracellular organelles (GO:0043229), membrane-bound organelles (GO:0043227), and intracellular membrane-bound organelles (GO:0043231) in the cellular component category (Fig. 5b, Online Resource S7). In addition to overlapping genome expression patterns, distinct and preferential genome expression was observed in the floral organs, indicating significant differences in key biological, molecular and cellular processes. During the floral transition (FB2), transcripts associated with RNA modification, cellular component assembly involved in morphogenesis, pollen wall assembly, regulation of cell growth and regulation of reproductive process were exclusively and abundantly expressed. Further size enlargement of floral buds (FB4) was associated with the exclusive expression of genes involved in cellular protein and macromolecule catabolic processes, the

regulation of multiorganism processes, responses to salicylic acid, responses to fungi, cellular nitrogen compounds, macromolecule biosynthetic processes, etc. Stamen development was accompanied by an enrichment of genes responsible for biological processes such as organelle assembly, photosynthesis, light harvesting in photosystem I, glucosinolate biosynthetic processes, S-glycoside biosynthetic processes, peptide metabolic processes, the light reactions, intracellular transport, cellular amide metabolic processes, glycosyl compound metabolic processes, external encapsulating structure organization, responses to lipids, oxidation–reduction processes, signal transduction, small signaling molecule metabolic processes, and cell communication. Similarly, CP development that controls female gametogenesis was regulated by enriched genes associated with the regulation of cellular protein metabolic processes, wax metabolic processes, fatty acid derivative metabolic processes, plant-type ovary

Fig. 5 Distribution of differentially expressed transcripts across chromosomes (**a**) and different families of transcription factors (**b**) identified from the floral tissues of *Brassica rapa*



development, carpel development, gynoecium development, floral whorl development, floral organ development, organelle organization, flower development, reproductive shoot system development, phyllome development, and shoot system development.

In the molecular function categories, genes responsible for antioxidant activity were enriched in FB2. Similarly, genes that exhibit kinase activity and ion binding activity, protein serine/threonine kinase activity and oxidoreductase activity (GO:0016709) were overrepresented in FB4 and STs. However, no exclusive molecular function categories were enriched with CP development. Both FB2 and FB4 were overrepresented with classes of genes whose products bind to regulatory regions of nucleic acids and exhibit peroxidase activity. In the case of cellular component synthesis, genes involved in the synthesis of the components of the thylakoid membrane protein complex, plastoglobules and photosystems; the thylakoid lumen; thylakoids; chloroplasts; and the nucleus were preferentially expressed in STs. Interestingly, in response to vernalization, cold-responsive genes (GO:0009409) were overrepresented in FB2, and the ST transcriptome suggests an impact of vernalization on floral transcription and male gametogenesis.

Functional classification of floral candidate genes, evolutionary conservation of gene expression and phylogenetic relationships

The protein sequence homology-based annotation with *A. thaliana* revealed 268 nonredundant, functionally known flowering genes in *B. rapa*, represented by 732 DEGs (Online Resource S1). Of these, 116 flowering genes were differentially expressed by fivefold in comparison with those in the CON (Fig. 6, Online Resource S8). Further classification based on their putative functions showed that 44 genes were related to flower development and meristem identity. Another 42 genes of the photoperiodism and temperature pathways, six genes of the vernalization pathway and 42 genes associated with the circadian clock and signaling were also identified. The vernalization-responsive genes AGAMOUS-LIKE 19 (Brapa_ESTC007968; expressed in FB2 and the CPs), MADS AFFECTING FLOWERING 4 (Brapa_ESTC013135; expressed in the CPs), FRIGIDA (Brapa_ESTC018555; expressed in the STs), and AGAMOUS-LIKE 19 (Brapa_ESTC018998 and Brapa_ESTC044364; expressed in the CPs) were identified. Among the hormone-responsive pathways, one positive regulator of flowering, Gibberellin-20-oxidase-3 (Brapa_ESTC029845), and three negative regulators, GAST1 protein homolog 5 (Brapa_ESTC007537), gibberellin 2-oxidase 8 (Brapa_ESTC024684) and gibberellin 2-oxidase 6 (Brapa_ESTC015242), were also identified (Online Resources S8 and S9). In general, 20 of the 27

annotated positive regulators were found to be upregulated, and seven genes that encode mainly phytochrome-interacting factors (CIB1 (Brapa_ESTC002825, Brapa_ESTC009500 and Brapa_ESTC044983), PIF5, PIF4, PIL6, and FD) were downregulated in all four floral tissues. Similarly, 24 of 51 predicted negative regulators (e.g., *SMZ*, *SPA3*, *CCA1*, *SVP*, *GLK2*, *GLK1*, *GA2ox6*, *TOE1*, *FLC*, *TEM2*, *LHY*, *SMZ*, and *CDF1*) were found to be downregulated in comparison with CON. Vernalization and other factors, such as the autonomous pathway, endogenous and exogenous stimuli, and photoperiod, expected to upregulate the positive regulators and suppress the negative regulators were detected in this study. Similarly, the expression of late flowering candidate genes involved in the aging pathway (*SMZ*, TARGET OF EARLY ACTIVATION TAGGED 1), photoperiodism (*SPA3*, CYCLING DOF FACTOR 1), the circadian clock (*CCA1*, LATE ELONGATED HYPOCOTYL), flowering time integration (*SVP*, FLOWERING LOCUS C), hormone pathways (GIBBERELLIN 2-OXIDASE 6, GIBBERELLIN 2-OXIDASE 8), and ambient temperature (TEMPRANILLO 2) as well as the expression of general candidates such as *GLK2* and *GLK1* was downregulated in this study.

Evidence in the literature suggests that evolutionary conservation of expression patterns is possible among closely related plant species (Andrés et al. 2015). To evaluate the conserved expression patterns of floral genes between *B. rapa* and *A. thaliana*, the expression pattern of 116 floral genes (FC=5) was compared with the expression pattern of Arabidopsis floral tissues [Apex3 (shoot apex after bolting)-ST-CP] from the microarray data retrieved by the AtGenExpress Visualization Tool (AVT, <http://jsp.weigelworld.org/expviz/expviz.jsp>) (Online Resource S10). The calculated expression correlation coefficient values showed that 60 of 116 flower gene candidates had a positive correlation (1.0–0.51), indicating a fair degree of conservation of floral gene expression patterns between Brassica and Arabidopsis species. The distinct expression pattern of Brassica is also evident, since 29 homologous genes from Arabidopsis had either negative (−0.50 to −1.0) or no correlation (0.1 to −0.19) with *B. rapa*.

Differential and preferential expression of transcription factors (FC ≥ 10) and transcriptional regulators in floral tissues

Transcription factors either activate or suppress floral organ development by interacting with specific target genes depending upon the physiological, environmental and genetic factors. In the present study, functional annotation of 6552 DEGs (FC ≥ 5) identified 382 TFs and 49 transcriptional regulators (TRs) from 3778 annotated genes via the iTAK tool. The predominantly expressed TFs (Online Resource S11) were AP2/ERFs (49 DEGs), followed by

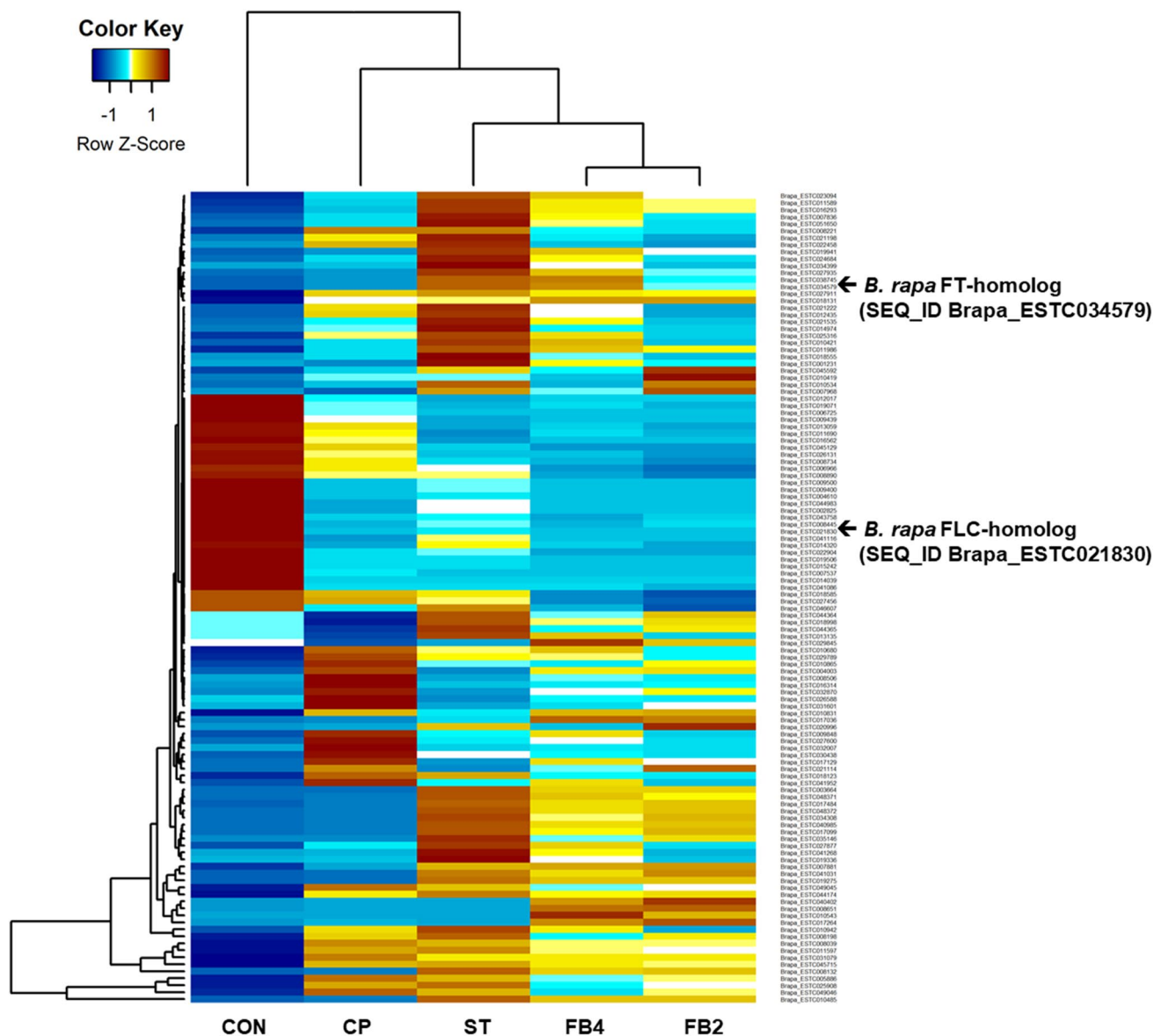


Fig. 6 Heatmap representing the expression profiles of 116 functionally annotated and differentially expressed (fold change ≥ 5) *Arabidopsis* genes homologous to those of *Brassica rapa*. A heatmap was generated using the heatmap.2 function provided by the R-package

gplots. Sample names and gene IDs are shown at the bottom and on the right of the heatmap, respectively. The fold change values were scaled per row (i.e., per unigene) to visualize the expression peaks of unigenes among samples

MADS-MIKCs (40 DEGs) and NACs (39 DEGs), while the most frequent TR was auxin responsive protein 28-like (AUX 28-like). Most of the TFs were commonly present in FB2, FB4, STs, and CPs, except for TFs such as CPP, AP2/ERF-AP2, TCP, ULT, S1Fa-like, HB-PHD, B3-ARF, NF-YC-9, and PLATZ, which were preferentially detected in STs and CPs, respectively. Additionally, 13 TFs showed preferential expression in early small flower buds (FB2), including overexpressed TFs, namely, LOB, bHLH, MYB, MADS-MIKC, and OFP TFs. The large flower buds (FB4) showed induced expression of C2C2-YABBY and MYB-related TFs. Similarly, 87 and 32 TFs showed preferential

expression in male gametophyte (ST) and female gametophyte (CP) genesis. Among the TFs whose expression fold change was no less than 10, MYB (Brapa_ESTC032430, Brapa_ESTC040454), bHLH (Brapa_ESTC018250) and LOB (Brapa_ESTC042519, Brapa_ESTC008313) TFs showed preferential expression in FB2. Similarly, NACs (Brapa_ESTC024914, Brapa_ESTC040921, Brapa_ESTC045884) and WRKYs (Brapa_ESTC045545, Brapa_ESTC043599) were dominant in the STs, while B3 (Brapa_ESTC008624), MYB (Brapa_ESTC021688) and GRF (Brapa_ESTC036641) TFs were associated with female reproduction organ development.

Identification of *B. rapa*-specific genes and their differential expression during flower development

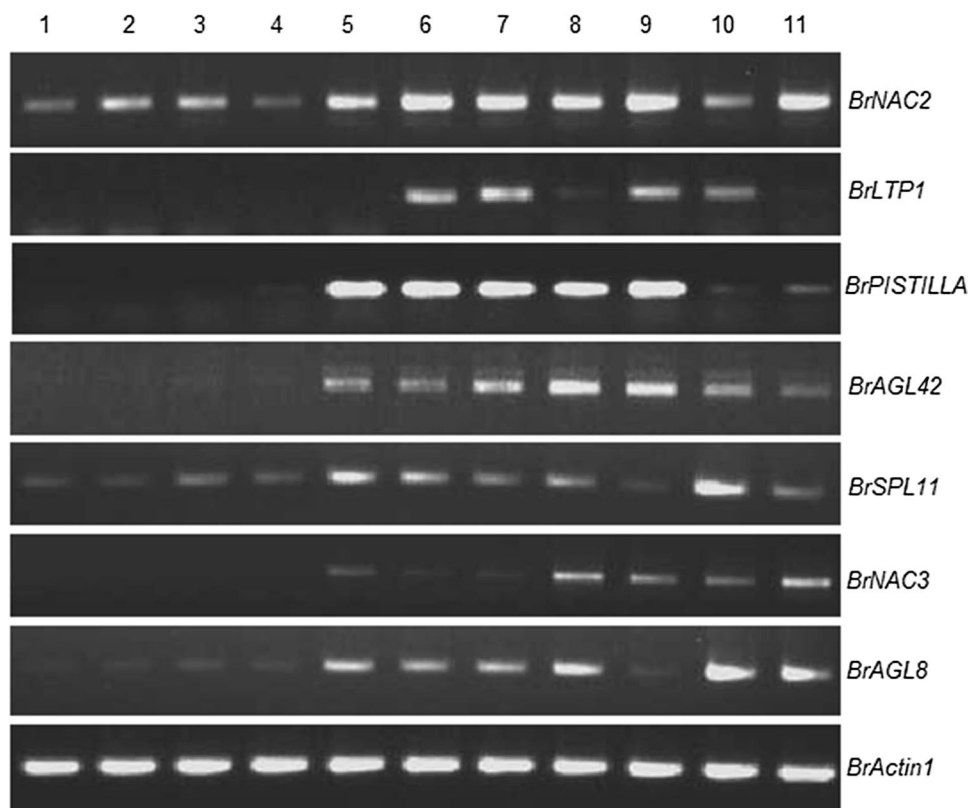
The functional annotation of the unigenes of Br300K revealed 744 unigenes with no sequence homology and/or annotation with *A. thaliana* gene models, indicating that they were specific to *B. rapa*. Interestingly, 210 unigenes were differentially expressed ($FC = 2$) during flower development, suggesting their possible involvement in that process (Online Resource S12). Of those DEGs, 131 presented at least a fivefold change in expression compared with those in vegetative tissues and were mostly present either in all floral tissues or exclusively in STs. Most of those DEGs (110) were overexpressed at any time of the floral organ development. Approximately 78 of 131 DEGs were uncharacterized proteins, and functional characterization of those DEGs will provide more insight into their role in flower development.

Spatiotemporal gene expression patterns of DEGs in floral and vegetative tissues

To confirm the floral tissue-specific expression of floral candidate genes identified from the microarray, seven DEGs were randomly selected (Online Resource S2), and their expression in floral tissues (FB2, FB4, ST, and CP) and vegetative leaves (CON) was assessed using semiquantitative RT-PCR (as mentioned in the materials and methods) (Fig. 7a). The RT-PCR

assay revealed that *PISTILLATA*, *LTP1*, *AGL42*, *NAC3*, and *AGL8* were abundantly expressed more in floral tissues rather than vegetative tissues, thus reaffirming the expression patterns predicted previously from the microarray. The expression of *NAC2* and *SPL11* was prominent in floral parts except for the CPs and STs, respectively. In addition, RT-PCR detected *NAC2* and *SPL11* and was used to quantify their expression in vegetative tissues including CON, VL5, LAV5, LAV10, OF and siliques; the magnitude of their expression was relatively lower than that in the flower buds. Interestingly, the effect of vernalization on the expression of *NAC2* was evident, as its expression was intensive in VL5 and LAV5 among vegetative tissues. However, *NAC3* (ortholog of AT3G15500.1) was expressed only in floral tissues and siliques, which indicates a functional difference between the two isoforms. *AGL8* (Brapa_ESTC017129), a flowering time and meristem identity gene, was preferentially expressed in FB2, FB4, and the CPs. Similarly, expression of *AGL42*, a key gene responsible for floral transition in *Arabidopsis*, was observed only in floral tissues and siliques and not in vegetative tissues. The expression of a lipid transfer protein (BrLTP1), which was previously known to regulate flower development in *B. napus* (Liu et al. 2014), was restricted to FB4, STs, and CPs. These results clearly indicate that genome expression is clearly restricted by developmental stage and other factors. Moreover, the expression patterns of those genes were consistent with the relative signal intensities predicted by the Br300K microarray.

Fig. 7 Reverse transcription (RT)-PCR assay of floral genes in different floral and vegetative tissues. The RT-PCR amplicons were resolved on a 1.3% agarose gel. Lane 1, leaves before cold treatment (CON); 2, leaves treated with cold for 5 days; 3, leaves from plants maintained at optimal conditions for 5 days after cold treatment; 4, leaves from plants maintained at optimal conditions for 10 days after cold treatment; 5, floral buds less than 2 mm in length (FB2); 6, floral buds 2–3 mm in length (FB4); 7, floral buds (length ≥ 4 mm; FB4); 8, open flowers; 9, stamens (ST); 10, carpels (CP); and 11, siliques



To understand how genomic collinearity is conserved in Brassica after whole-genome duplication and subsequent fractionation events, we investigated the evolutionary relationship of some key flowering genes (*AGL42*, *NAC3*, *LTP1*, *WUS*, *SEP2*, *SPL11*, *PISTILLATA* and *APETALA*) in this study. Phylogenetic analysis (Online Resource Fig. 1) showed that the *AGL42*, *PISTILLATA* and *WUS* (Online Resource Fig. 3a, e, h) gene groups of *B. rapa* share a closer relationship at the protein sequence level with those of *B. campestris* than with those of *B. oleracea*, *B. napus*, *B. juncea*, and other Brassicaceae members while *SPL11* share more similarity with *B. napus*. Nonetheless, the relationships of the *APETALA*, *LTP1*, *SEP2*, *NAC2* and *WUS* gene groups with those of other *Brassica* spp. are not conclusive, as indicated by their bootstrap values. However, most of the floral genes of *Brassica* spp. have a closer relationship with those of Brassica rather than those of *Arabidopsis*, suggesting the impact of genome duplication and fractionation events and their evolutionary selection pressures in Brassica.

Localization and detection of the expression of *BrLTP1* in floral tissues using in situ hybridization

Plant lipid transfer proteins known to possess multiple functions and, recently, *BrLTP1*, have been shown to play a role in epicuticular wax deposition, cell proliferation and flower development (Liu et al. 2014). However, their expression patterns during flowering and organ specificity remain unclear. In situ hybridization of the mRNA of any target gene can provide direct evidence of the localization and spatiotemporal expression pattern of candidate genes in plant tissues. In the present study, in situ hybridization of *BrLTP1* (*Brapa_ESTC001844*) transcripts by complementary DIG-labeled sense (Fig. 1a and c) and antisense probes (Fig. 1b and d) revealed that their expression was localized to stamen (ST), pollen sac (PA), pistil (PI) and carpel (CP) tissues. Strong expression was observed in the pollen sac and/or pollen. *BrLTP1* activity was also found in other parts of floral buds, but was expressed at relatively low levels. This result indicates that *LTP1* might participate in the development of male and female reproduction organ development as part of flower development in *B. rapa*. This finding is consistent with the relative signal intensities detected with the Br300K microarray and semiquantitative RT-PCR (Fig. 7a), supporting that the estimates of gene expression in the present study are reliable.

Analysis of the spatiotemporal expression of *LTP* in the floral tissues of *B. rapa* transgenic reporter lines (*BrLTP* promoter::*GUS*)

Floral organ-specific genes are important for understanding flower development and genetic manipulation to enhance

yield traits. Therefore, we also attempted to study the putative function of *BrLTP* based on its promoter activity. To check the tissue specificity of the floral candidate gene *LTP* and to identify its putative functional role in flower development, the promoter region (1.537 kb upstream of the start codon) of *LTP* was fused to a reporter gene, β -glucuronidase (*GUS*), and then genetically transformed into *B. rapa* (Fig. 2). The floral tissues of the transgenic reporter lines were subjected to a GUS staining assay, and the results showed that *BrLTP* promoter-driven *GUS* expression was mainly observed in the STs (Fig. 2b). GUS activity was also found in the carpel tissues. In addition, activity was found in the tips of early small (FB2) and large flower buds, but their expression was low in comparison with the expression in the ST and CP tissues. There was no expression in the sepals, receptacles, the stalk of the flower or stem, indicating that *BrLTP* activity is likely restricted to stamen and carpel development during floral transition. This result further supports the expression pattern identified in the Br300K array, where prominent expression of *LTP* was reported in the STs followed by CPs and where expression was relatively low in FB2 and FB4.

Discussion

Understanding the floral transition and flower development, including that of male and female organs, in *B. rapa* is important for genetic manipulation to develop novel lines, such as late-bolting *B. rapa*, temperature-resilient and male-sterile lines. However, previous studies with Brassica species have been limited to flowering time, male sterility and vernalization response pathways; no system-wide attempts have been made to determine the continuous series of gene expression responsible for the floral transition to maturation, including gametogenesis. Therefore, to define the subset of genes as well as their relative abundance and functions in flower development, we compared the transcriptome profiles of floral buds from the early (FB2 and FB4) and late developmental stages (STs and CPs) with the transcriptome of leaves. In this study, 30% of the genome of *B. rapa* was accounted for by 6848 gene models as flowering genes. Among the differentially expressed transcripts, 13.8% had a fivefold change in genome expression, indicating their true involvement in flower development. Interestingly, 70% of those flowering genes were upregulated. Among the floral tissues, a higher degree of transcriptional reprogramming occurred in the STs, suggesting a major shift in transcriptional reprogramming from the general floral transition to male reproduction organ development followed by FB4, FB2, and CPs. A subset of 1536 DEGs may be involved in key cellular and metabolic processes associated with floral transition and development, as evident by their expression

in all floral tissues. Additionally, we reported that 236 and 49 DEGs were preferentially expressed in FB2 and FB4. Similarly, 1723 and 232 DEGs were detected in ST and CP development. The initial floral transition (FB2) was accompanied by transcriptional reprogramming by 2501 upregulated and 1138 downregulated DEGs. The subsequent flower development has both overlapping and stage-specific DEGs according to their organ relatedness. As expected, the highest numbers of overlapping DEGs were observed between the developmental stages of FB2 and FB4 compared with the STs and CPs. Further analysis revealed that 1094 overlapping DEGs across four developmental stages were upregulated in comparison with the CON; these DEGs could be considered potential floral markers for *B. rapa*. Moreover, FB2 and FB4 share a large number of downregulated DEGs that are presumed to be responsible for vegetative tissue development rather than floral tissue development. These results indicate that each organ had a defining genome expression pattern and that the related organs had relatedness in terms of genome expression. Most of those DEGs originated from chromosomes 3 and 9 of *B. rapa*. A previous study found that most of the flowering time genes were distributed on chromosome 9 of *B. rapa*, while they were distributed on chromosome C03 of *B. napus* and *B. oleracea* (Li et al. 2018). Similarly, Wu et al. (2018) found that the highest number of MADS genes were distributed on chromosomes 3 and 9 of a recent *B. napus* interspecific hybrid of *B. rapa* and *B. oleracea*. Incidentally, chromosome 9 of Brassica species harbors the genes associated with pathogen resistance (Fu et al. 2019) and glutathione transferase genes, indicating the importance of chromosomes 3 and 9 of Brassica species in development and stress responses.

The knowledge of organ-specific expression patterns is crucial because it corresponds to their function in flower development. The analysis of the genome expression of each organ provided some interesting facts on key biological, molecular and cellular processes regulating the floral transition, male and female reproductive organogenesis, and maturation. Gene ontology terms indicated that the genetic makeup of flower development was most responsive to the stimuli (GO:0050896) and stress (GO:0006950). Kazan and Lyons (2016) reported that epigenetic factors, including biotic and abiotic stress conditions, altered flowering time as part of the stress adaptation strategy. Further, genes functioning in enzymatic activities such as catalytic, oxidoreductase and transcriptional regulation were most frequently expressed during flower development. The functional enrichment of preferential genome expression in each floral organ suggested a significant difference in key processes that control flowering. Early floral transition induces activities associated with RNA modification, the assembly of cellular components involved in morphogenesis, pollen wall assembly, the regulation of cell growth and the regulation

of reproductive processes. Interestingly, ST development was accompanied by the enrichment of genes associated with photosynthesis, and CP development was promoted by enriched genes associated with the regulation of protein metabolic processes, wax and fatty acid derivative metabolic processes, ovary development, carpel development, gynoecium development, floral whorl development, etc. The enrichment of photosynthesis- and chloroplast component synthesis-related DEGs in the STs suggests that stamens are photosynthetically active organs; however, their contribution has been reported to vary among plant species (Brazel and Ó'Maoiléidigh 2019). Previously, a rice stamen identity gene and a MADS-box protein were reported to bind and reduce photosynthesis-associated proteins during stamen development (Chen et al. 2015). In contrast, in the present study, in comparison with leaf transcriptome expression, the expression of these genes was upregulated in stamen development. Photosynthesis-related genes have also been reported to be enriched in lotus floral organs such as stamens, stamen petaloids and petals (Lin et al. 2018). Interestingly, in response to vernalization, cold-responsive genes (GO:0009409) were overrepresented in FB2, and the ST transcriptome suggests the impact of vernalization on floral transcription and male gametogenesis.

Flowering is an intricate and complex process involving a network of genes and regulatory elements that are fairly conserved among plant species. Therefore, a comparison of the flowering pathways of closely related model species may provide important clues about the evolutionary conservation of gene function (Vining et al. 2015). Previous reports have shown that *B. rapa* and *Arabidopsis* have a close syntenic relationship at the chromosome and gene level (Tang and Lyons 2012). Further, in the present study, we observed that 85% of the unigenes of the Br300K microarray were homologous to 18,295 *Arabidopsis* genes, indicating close similarity even after whole-genome triplication of *B. rapa* (Tang et al. 2012; Wang et al. 2015). Our study identified 268 flowering gene orthologs from *Arabidopsis*, of which 116 were classified based on their putative role in flower development. This classification includes gene classes of both positive and negative regulators of flower meristem identity, photoperiodism, the ambient temperature pathway, the circadian clock, the vernalization pathway, aging, and the gibberellin pathway. To date, 306 genes participating in four major pathways, including those involving the photoperiod, the circadian clock, autonomy, aging, ambient temperature, and gibberellins, have been identified (Bouché et al. 2015). Most of the known positive regulators were found to be induced, while the expression of repressor/negative regulators was reduced in response to vernalization. Notably, the late flowering genes of the aging, photoperiodism, circadian clock, flowering time integration, gibberellin response and temperature pathways being downregulated in this study

corroborates the findings of Song et al. (2016), who reported that downregulation of floral repressors is essential for early flowering in *B. rapa*.

The *B. rapa* and *A. thaliana* genomes conserve a fair amount of positive expression patterns among the reported 116 floral genes and are expected to have functional similarity (Andrés et al. 2015). For instance, the vernalization pathway, which suppresses the FLC floral repressor genes in *B. rapa*, was found to have a positive correlation [coefficient (0.99)] with *A. thaliana*, suggesting that vernalization-mediated pathways have been conserved between those species (Xiao et al. 2013; Andrés et al. 2015). Nonetheless, it is also evident that some of the floral genes of *B. rapa* have a negative correlation for expression, possibly due to evolutionary selection pressure in the form of fractionation just after gene duplication events occurred in Brassica species (Sankoff et al. 2010). We reported 210 *B. rapa*-specific unigenes, of which approximately 110 were overexpressed by fivefold in comparison with those in the CON. Unfortunately, most of these DEGs warrant functional characterization, which is expected to provide more insight into their role in flower development. Previous reports also provided evidence that some portion of gene regulatory switches that control male fertility (the presence of self-incompatible genes) and the vernalization response in *B. rapa* were novel and absent in Arabidopsis (Dong et al. 2013). This evidence indicates the existence of a distinct flowering regulatory mechanism apart from several flowering homologs of Arabidopsis found in *B. rapa* (Online Resource S10).

Differential expression of transcription factors and their functions in flower development

Transcription factors directly regulate the expression of specific target genes according to the genetic composition and their interaction with various signals. Previous studies have revealed that TFs can influence several pathways involved in the transition from the vegetative to reproductive phase, such as the biosynthesis of anthocyanins (R2R3-MYB, basic helix–loop–helix and WD40 types), sepal formation (class A and MADS-box TFs), petal formation (class A, B and E and MADS-box TFs), stamen formation (class B, C and E TFs), carpel formation (class C and E TFs), ovule formation (class C, D and E TFs), gynoecium development (AP2/ERFs), floral meristem initiation, floral organ identity, organ growth, and gynoecium patterning (AIL/PLT TFs) (He et al. 2010; Krizek 2011; Theissen et al. 2016; Durán-Medina et al. 2017). In the present study, we identified 382 TFs ($FC \geq 5$) along with 49 transcriptional regulators (TRs) (Online Resource S11). The predominantly expressed TFs were AP2/ERF, MADS-MIKC, and NAC TFs, while the most frequent TR was auxin responsive protein 28-like (AUX 28-like). Although TFs were commonly present in

FB2, FB4, STs, and CPs, some TFs showed preferential expression in specific floral tissues, suggesting a differential regulatory mechanism for each stage of flower development (Zhang et al. 2015).

The three highly abundant ($FC \geq 10$) NAC TFs (Brapa_ESTC024914, Brapa_ESTC045884, and Brapa_ESTC040921) of the STs in *B. rapa* are expected to modulate various processes, including anther dehiscence (Shih et al. 2014), cell division and development of the shoot apical meristem (Shao et al. 2015). However, identification of their potential target genes is essential to understand their exact role in flower development. These TFs are important targets for future studies associated with flower development. While most of these TFs' functional role in flower development has yet to be identified, MADS-MIKC and NAC TFs have been widely studied in crop models. The MADS-MIKC transcription factor regulates floral organ identity and is a part of the main component of ABC floral model development (Heijmans et al. 2012). The functional role of MADS-MIKC in flower development has diversified during evolution and plays a critical role in photoperiodism, including flowering time integration, meristem identity, and vernalization. TFs such as CPP, AP2/ERF-AP2, TCP, ULT, S1Fa-like, HB-PHD, B3-ARF, NF-YC-9, and PLATZ were preferentially observed in the STs and CPs. Similar to other TFs, nuclear factor (NF-YC) TFs are involved in multiple cellular and developmental processes, including photoperiod-dependent flowering time, gametogenesis, embryogenesis, seed production, ABA signaling and drought resistance (Fornari et al. 2013). The interaction of NF-YC9 with NF-YB2 and NF-YB3 is required for flowering; moreover, NF-YC9 interacts with CONSTANS (CO) for CO-mediated floral promotion in Arabidopsis (Kumimoto et al. 2010). The functional role of most other TFs is complex and elusive because TFs tend to interact with multiple targets depending upon the plant response to signals. Similar to TFs, the most predominant TR, AUX-28-like, regulates flower initiation, flower identity and patterning by responding to auxin hormones and its associated gene expression (Krizek 2011). The transcriptional regulators identified in this study warrant in-depth studies to understand their role in flower development.

Preferential gene expression patterns of floral DEGs in FB2, FB4, ST, and CP development

The majority of the positive regulators of flower development among the 116 Arabidopsis homologous genes reported in this study (Online Resource S8) were induced mainly in response to vernalization, the autonomous pathway, and/or by other unknown factors as revealed previously in other species (Li et al. 2015; Kazan and Lyons 2016; Wang et al. 2017). Interestingly, among the downregulated positive regulator genes, PHYTOCHROME-INTERACTING

FACTORS are prominent candidates because they are known to participate in the repression of plant responses to light, another critical regulator of flower development. However, the function of PIFs is known to be influenced by species (Jeong and Choi 2013); hence, their role in flower development in *B. rapa* needs further analysis. The importance of vernalization is evident in this study, where 24 of 51 known negative regulators were reduced during flower development, as stated by Amasino (2005). Among the negative regulators, *GA2ox6*, which is known to convert bioactive gibberellin into its inactive form, was downregulated to initiate early flowering (Hu et al. 2017). *FRIGIDA* is known to promote the expression of the floral repressor *FLC*, which is downregulated in response to vernalization and/or the autonomous pathway and, in turn, accelerates flowering (Shindo et al. 2005). *Arabidopsis FLOWERING LOCUS T (AtFT)* is usually expressed in leaf vasculature and transported to the shoot apical meristem. This floral integrator gene interacts with *FLC* and *CONSTANS (CO)* (positive regulator of flowering) to determine the timing of flowering (Kazan and Lyons 2016). However, in the present study, the expression of *BrFT* (SEQ_ID Brapa_ESTC034579) was relatively higher in FB2 and FB4 than in the CON; further, the expression of *BrFT* peaked in the STs (FC=20) before decreasing in the CPs, suggesting contrasting expression with that of *AtFT*.

Subsequent analysis of genome expression revealed 12 stress-responsive DEGs (FC=5 or more) in floral tissues. Importantly, *ERD4* (a drought tolerance gene) and a salt-responsive unknown protein (AT1G16850) were upregulated as part of the transcriptomic changes that control male gametophyte (ST) development. Similarly, cold stress-responsive (Brapa_ESTC048390 and Brapa_ESTC003818) and oxidative stress-responsive gene expression was also detected in the floral organs. While a direct link with flower development has yet to be identified, their role in flower development seems to be critical, as evidenced by their expression pattern. Understanding the transcriptional regulatory mechanism of stamen development would facilitate novel approaches for the production of improved crops (Chen et al. 2015). In our study, a relatively higher degree of transcriptional reprogramming occurred in the STs (5628 of 6552 DEGs) than in the other floral tissues, which included 1723 ST-preferential DEGs. Of these, 877 DEGs were upregulated by fivefold, while 846 were downregulated. Most of these DEGs lacked a functional annotation and, hence, were considered novel floral transcripts. Polygalacturonase 4 (Brapa_ESTC009221), Ralf-like 9 (RALFL9) (Brapa_ESTC048515), RALFL4 (Brapa_ESTC007624), SKU5 similar 11 (Brapa_ESTC001190), an SNF2-type transcriptional regulator (Brapa_ESTC027126) and a bZIP family transcription factor (Brapa_ESTC009320) were some of the highly upregulated transcripts found in the STs. Among

the DEGs preferentially expressed in the STs, polygalacturonase 4 (Brapa_ESTC009221) functions in the formation and release of pollen grains in *B. rapa* (Futamura et al. 2000). Similarly, the pollen-specific RALF gene regulates pollen tube elongation in Brassica, similar to tomato RALF (Covey et al. 2010). Futamura et al. (2000) reported exclusive expression of polygalacturonase present in the male flowers of dioecious willow (*Salix gilgiana* Seemen). Further, genes associated with photosynthesis and chloroplast component synthesis were preferentially expressed in STs (Online Resource S7)

Analysis of the spatiotemporal gene expression patterns of floral DEGs using RT-PCR, in situ hybridization and transgenic reporter lines

The spatiotemporal gene expression pattern of DEGs predicted in the Br300K microarray was validated with selected candidates using RT-PCR, in situ hybridization and transgenic reporter lines in this study. For this purpose, 7 DEGs representing the floral transition (*AGL42*), flowering time and meristem identity (*AGL8*), reproductive and vegetative shoot development (*SPL11*), organ-specific expression (*PISTILLATA*), flowering-specific expression (*LTP1*, *NAC3*) and all stage of expression (*NAC2*) were selected. The RT-PCR assay revealed that the *LTP1*, *AGL42*, *NAC3*, and *AGL8* genes were abundantly expressed in FB2, FB4, STs and CPs compared with vegetative tissues. Expression of *NAC2* and *SPL11* was detected in vegetative (CON, VL5, LAV5, and LAV10) and flowering stages (FB2, FB4, STs, and CPs), although the expression in floral tissues was prominent. Our study found that *NAC2* is a vernalization-responsive gene, as evident from the intensive expression pattern observed in VL5 and LAV5. *BrLTP1* was restricted from FB4 to ST and CP development, as reported previously by Liu et al. (2014). As shown by the microarray, *BrPISTILLATA* showed preferential expression in FB2, FB4, and the STs compared with the CPs and CON. As mentioned previously, *PISTILLATA* is exclusively expressed in STs (Sobral and Costa 2017) and is known to be involved in ST development (Bowman et al. 1989; Honma and Goto, 2000). The preferential expression of *PISTILLATA* in STs suggests evolutionary conservation of floral genes across species (Vining et al. 2015). *AGL42* encodes a MADS-box TF involved in the floral transition in *Arabidopsis* (Dorca-Fornell et al. 2011) and promotes flowering at the shoot apical and axillary meristems, which depends on the gibberellin pathway. *SPL11* regulates root, shoot and flower development (Xu et al. 2016), which was supported by their expression pattern in this study. Similarly, in addition to its demonstrated role in ovule integuments (Vining et al. 2015), *NAC2* participates in the signaling pathways of ethylene and auxin, the salt stress response and lateral root

development (He et al. 2010). These results clearly indicate that genome expression is clearly restricted by developmental stage and other factors. Moreover, the expression pattern of those genes was consistent with the relative signal intensities predicted with the Br300K microarray, indicating that spatially limited expression of genes is crucial for flower development. Plant LTPs play multiple roles, including those in epicuticular wax deposition, cell proliferation and flower development (Liu et al. 2014). However, their expression patterns during flowering and organ specificity remain unclear. In the present study, in situ hybridization of *BrLTP1* (Brapa_ESTC001844) transcripts revealed strong expression in the stamens (STs), pollen sac (PA), pistil (PI) and carpels (CPs). This result indicates that *LTP1* might participate in the development of male and female reproduction organ development as part of flower development in *B. rapa*. This finding is also consistent with the relative signal intensities detected with the Br300K microarray and semi-quantitative RT-PCR (Fig. 7a), supporting that estimates of gene expression in the present study are reliable. To prove their involvement in male and female reproduction organ development in *B. rapa* further, we developed transgenic reporter lines (Fig. 2), which could be useful for enhancing our understanding of the role of these genes in flower development. The GUS staining assay revealed BrLTP promoter activity in the STs and CPs and relatively little activity in the tips of early small (FB2) and large flower buds; moreover, no activity was detected in other parts of the flowers, such as the sepals, receptacle, stalk of the flower or stem. These results provide strong direct evidence that *BrLTP* is crucial for the development of stamens and carpels and therefore likely plays a role in seed development (Dong et al. 2004). Our results support the findings of Liu et al. (2014), who reported detectable phenotypic changes in the stamens and carpels of BrLTP1-overexpressing *B. napus* lines. Moreover, our study provided evidence that genomic collinearity is conserved among key flowering genes in Brassica species even after whole-genome duplication and subsequent fractionation events (Fig. 7b). The protein sequence-based analysis also showed that flowering genes, including PISTILLATA, had the highest similarity with those of *A. thaliana* rather than with those of other Brassica species, indicating different degrees of fractionation and evolutionary selection pressures across the genomes of Brassica species rather between Arabidopsis.

In conclusion, our study identified 6848 flowering genes, including several Arabidopsis homologs, 210 Brassica-specific genes and 382 genes that encode TFs. The identified flowering genes are involved in a network of six regulatory pathways, and their preferential expression will reveal the potential molecular mechanism responsible for the floral transition and stamen and carpel development in vernalization-sensitive Brassica species. It is apparent that genome

expression in FB2, FB4, STs and CPs overlapped but significantly differed from that in vegetative tissues, suggesting that developmentally related organs can exhibit a similar pattern of gene expression. According to the expression pattern among floral organs, CP development is least dissimilar to that of vegetative tissues, and relatively more transcriptomic changes were observed with stamen development. Although the role of many genes, including those encoding TFs and TRs and *B. rapa*-specific genes, remains an open avenue and requires further analysis, the datasets and the transcriptomic information provided here will significantly contribute to future research.

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Author contributions SL, JK and MJ conceived and designed the study. MM and MN conducted the bioinformatics analysis, analyzed the data and drafted the manuscript. MM, and SL submitted the array data to the NCBI Gene Expression Omnibus database. SL, JK, JH, ML and MJ revised the manuscript. All the authors agreed on the contents of the paper.

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Compliance with ethical standards

Conflict of interest Soo In Lee declares that he has no conflict of interest. Muthusamy Muthusamy declares that he has no conflict of interest. Muhammad Amjad Nawaz declares that he has no conflict of interest. Joon Ki Hong declares that he has no conflict of interest. Myung-Ho Lim declares that he has no conflict of interest. Jin A Kim declares that she has no conflict of interest. Mi-Jeong Jeong declares that she has no conflict of interest.

Data availability statement The complete set of genome-wide expression data from this study was submitted to the Gene Expression Omnibus database (<https://www.ncbi.nlm.nih.gov/geo/>) under accession number GSE128989, and the necessary information was also submitted as supplementary data in this article.

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